Bell Quantum Bound 2√2 from Spinor Bundle Holonomy

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Abstract

The Bell-CHSH quantum bound $S = 2\sqrt{2} \approx 2.828$ (Tsirelson) emerges in UQFF from the holonomy of the spinor bundle SO(26) Clifford module across compactified dimensions. Quantum non-locality is therefore a geometric phenomenon — not a "spooky action" — and arises from the topology of the 26D compactification.

Part 1: The Bell Bound

Statement

For any classical theory with local hidden variables: $S \leq 2$ (CHSH). Quantum mechanics: $S_{max} = 2\sqrt{2} = 2.828$ (Tsirelson 1980). Observed (Aspect 1982, Hensen 2015): $S \approx 2.81$ violating Bell.

UQFF interpretation

$S = 2\sqrt{2}$ emerges as the spinor bundle holonomy parameter when partial measurement on entangled pair is rotated through the SO(26) compactified holonomy.

Part 2: Why 2√2 Specifically

The Clifford module spinor dimension in UQFF 26D compactification is $2^{(D_{crit}/2)} = 2^{13} = 8192$. The minimal entangled state has support on 4 modes, and the maximal correlation across a partial measurement rotation θ is $\sqrt{2} \cdot \cos(\theta - \pi/4)$. The CHSH sum then gives maximum $2\sqrt{2}$ at $\theta = \pi/4$.

Part 3: Experimental Tests

- **Aspect 1982**: $S \approx 2.6 \pm 0.2$ (photon polarization)
- **Hensen 2015**: $S \approx 2.42$, loophole-free (NV centers)
- **Tsirelson bound**: $S \leq 2\sqrt{2} \approx 2.828$ strict

Part 4: Why Not S = 4 (Algebraic Maximum)?

The PR-box (Popescu-Rohrlich) allows $S = 4$ in principle, but UQFF predicts $S \leq 2\sqrt{2}$ because the spinor bundle is a finite-dimensional representation. Going beyond $2\sqrt{2}$ would require an infinite-dimensional Hilbert space which UQFF excludes via the 26! finite bound.

Conclusion

Bell quantum non-locality is the geometric signature of UQFF spinor bundle holonomy across the 22 compactified dimensions. The Tsirelson bound $2\sqrt{2}$ is the spinor representation's maximum correlation parameter.

Framework Version: UQFF 5.27+